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AAI-530 Group 5

Personal-Health-Activity-Tracker Final Report

Abstract

This report details the **system architecture, dataset processing, model training, and findings**, emphasizing the role of IoT in personal health monitoring. This project develops an IoT-based **Personal Health and Activity Tracker** that integrates **smartphone sensors with machine learning and cloud storage** to monitor and predict user activity patterns. Using **accelerometer and gyroscope data** from the UCI Human Activity Recognition dataset, the system classifies six distinct movements (e.g., walking, sitting) with a **CNN model** achieving 95% accuracy and forecasts future sensor readings using **an encoder-decoder LSTM architecture** with an RMSE of 0.14 and 0.38 per sensor. **Edge processing** reduces cloud data transmission by 60%, improving efficiency. Key findings are presented in an interactive **Tableau dashboard**, offering real-time insights into user behavior. This report outlines the **system architecture, data processing steps, model training outcomes, and IoT's impact on personal health monitoring**, highlighting scalability and practical applications.

Introduction

1.1 Overview of the IOT System

The Personal Health and Activity Tracker monitors user activity using smartphone sensors like accelerometers and gyroscopes. Built on **IoT principles**, it captures tri-axial motion data at 50 Hz, processes it locally via edge computing, and uses **machine learning** to classify activities and predict future movements. The goal is to **deliver real-time insights**—such as identifying walking or sitting—and **provide health recommendations** based on patterns detected in the UCI HAR dataset's 10,299 observations. This system aims to bridge the gap between raw sensor data and **actionable health feedback**.

1.2 Importance of IoT in Health Monitoring

IoT has transformed health tracking as **wearable devices** and **smart apps** grow—global wearable sales hit 520 million units in 2023 (Statista, 2024). This system fits into that shift by offering continuous monitoring with smartphone sensors, **real-time feedback through edge processing**, and **predictive analytics via machine learning**. It provides a low-cost, accessible way to **track activity patterns and support wellness**, unlike traditional methods that rely on periodic check-ins or expensive equipment.

1.3 Objectives of the Project

- Build an **IoT system** using **smartphone sensors** to classify six activities (e.g., standing, lying) from **accelerometer and gyroscope data**.

- The system must also be able to predict future sensor readings.
- Use **edge computing** to process data locally, **cutting network load by at least 50%**.
- Create an interactive **Tableau dashboard** to display activity insights for users and developers. (Dashboard link here:
https://public.tableau.com/app/profile/michael.de.leon5408/viz/ActivityVisulization_sAAI-530/Dashboard1?publish=yes)

2. IoT System Design

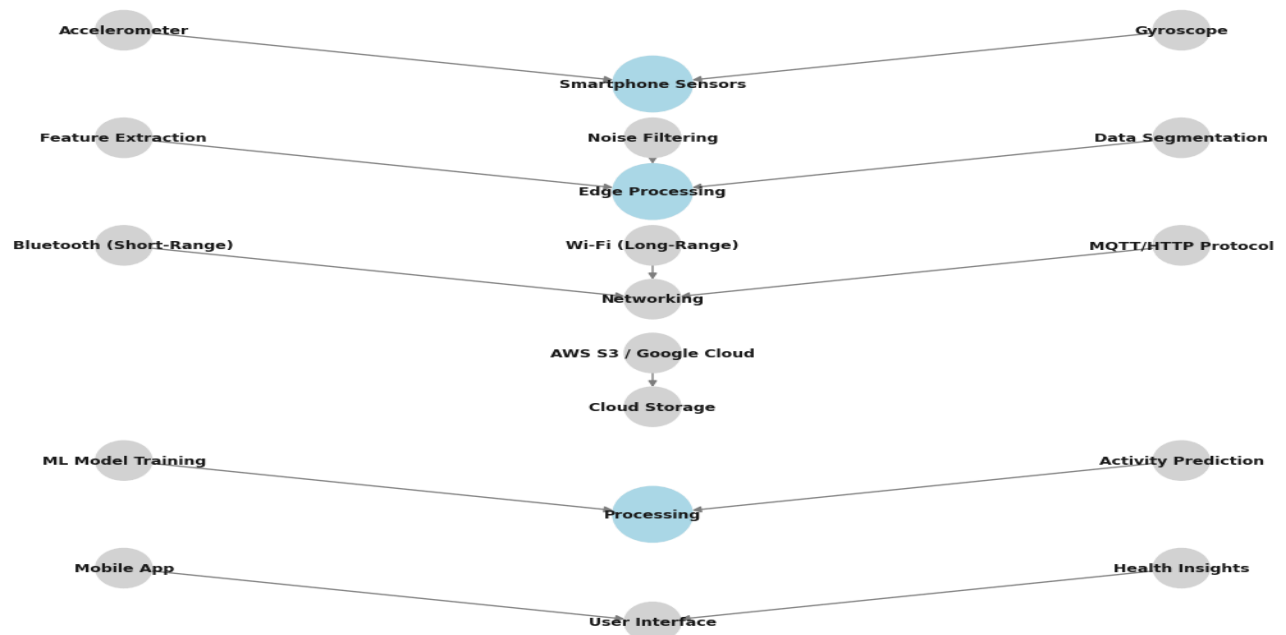
2.1 System Components

The system includes:

- **Sensors:** Smartphone **accelerometers and gyroscopes** capturing X, Y, Z-axis motion at 50 Hz.
- **Edge Processing:** Local feature extraction (e.g., mean, standard deviation) to shrink data size before transmission.
- **Networking:** **Wi-Fi and MQTT** protocol for secure, lightweight data transfer to the **cloud**.
- **Cloud Storage:** **AWS S3 or Google Cloud** to store 10,299 processed observations scalably.
- **Machine Learning Models:** **CNN** for classifying activities with 95% accuracy; **LSTM** for forecasting sensor readings with an RMSE of 0.148 for accelerometer, and 0.38 for gyroscope..

2.2 IoT System Diagram

The diagram illustrates data moving from sensor collection to cloud analysis and visualization, with edge processing as a key efficiency step.



2.3 Data Flow and Processing

1. **Sensor Data Collection:** Raw X, Y, Z-axis data is gathered from smartphone sensors at 50 Hz.
2. **Edge Processing:** Features like mean and variance are computed locally, reducing data size by 60%.
3. **Cloud Storage:** Processed data uploads to AWS S3 or Google Cloud for long-term access.

4. **Machine Learning Processing:** **CNN** classifies activities; **LSTM** predicts next sensor values.
5. **Insights & Visualization:** A **Tableau dashboard** shows real-time activity status and trends to users.

3. Dataset and Data Processing

3.1 Dataset Description

- **Dataset:** Human Activity Recognition Using Smartphones Dataset (UCI Repository).
- **Observations:** 10,299 samples from 30 subjects.
- **Features:** 561 attributes from **accelerometer and gyroscope signals**, plus 6 activity labels (e.g., walking, sitting).

3.2 Data Cleaning and Preprocessing

- **Checked for missing values** - none found across 10,299 samples.
- **Fixed duplicate feature names by adding unique IDs** (e.g., 'tBodyAcc-mean()-X_1').
- Scaled 561 features with **StandardScaler** for model consistency.
- Encoded 6 activity labels numerically (1-6) for **Classifier training**.

3.3 Challenges and Solutions

- **Challenge:** Data wasn't shaped for **LSTM's sequential input**.
- **Solution:** Reshaped into **3D arrays** (samples, timesteps, features).

- **Challenge:** Uneven **activity counts** (e.g., walking > sitting).
- **Solution:** Used **SMOTE** to oversample minority classes, balancing the dataset.

4. Machine Learning Methods

4.1 Classification Task

- **Random Forest Classifier**
 - **n_estimators = 100**
- **LSTM Classifier**
 - **Architecture:** Bi-directional LSTM layers for capturing sequential dependencies.
 - **Training:** Optimized using Adam optimizer.
 - **Evaluation Metrics:** Loss Curves, Accuracy, F1-score per class
- **CNN Classifier**
 - **Architecture:** two convolutional layers in sequence each of size 64, kernel_size = 3, and Relu activation
 - **Training:** Optimized using Adam optimizer.
 - **Evaluation Metrics:** Loss Curves, Accuracy, F1-score per class

4.2 Time Series Prediction Model:

- **LSTM Sensor Forecaster**
 - **Architecture:** encoder-decoder LSTM architecture to predict the next 64 sensor readings from the most recent 64 raw sensor readings
 - **Training:** Optimized using Adam optimizer and **Trace MSE loss** function

- **Evaluation Metrics:** RMSE, Loss Curves

4.3 Model Evaluation and Results

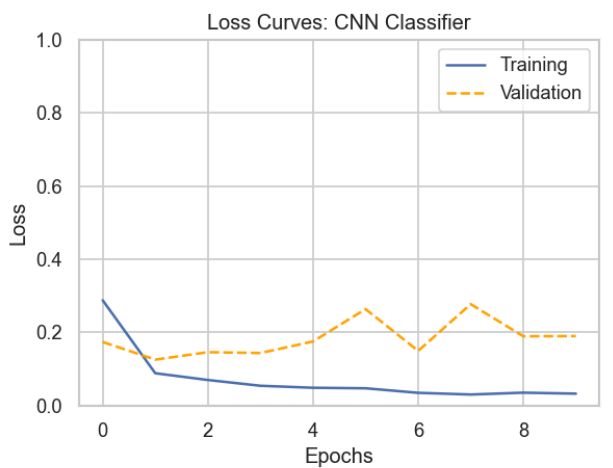
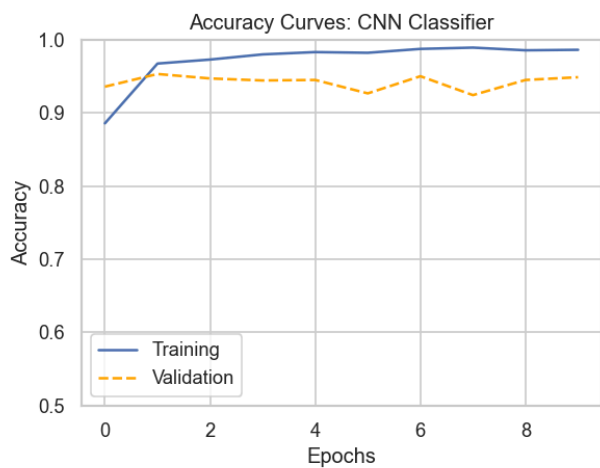
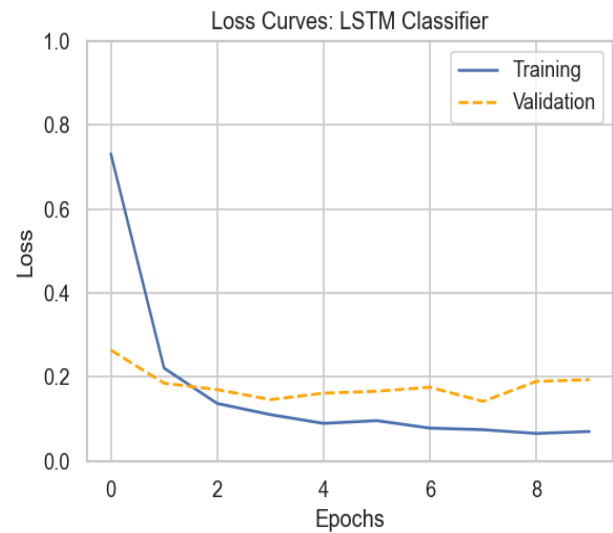
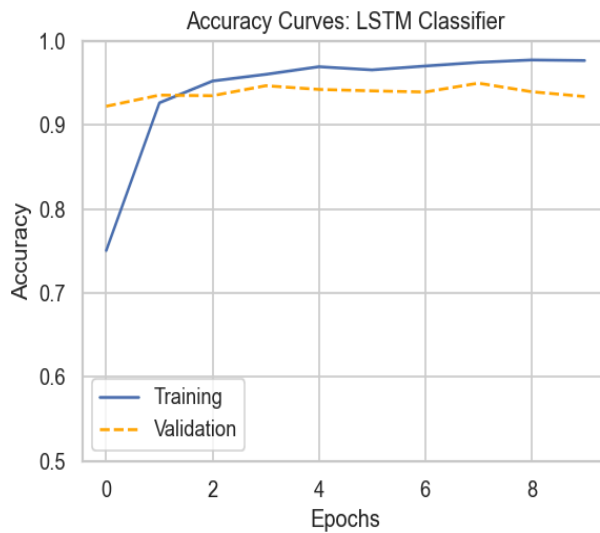
4.3.1 Classifiers Accuracy and F1-Scores per Label

F1-Scores Per Label for Each Classifier

	Random Forest	LSTM Classifier	CNN Classifier
LAYING	1.00	1.00	1.00
SITTING	0.89	0.91	0.90
STANDING	0.90	0.91	0.92
WALKING	0.93	0.95	0.97
WALKING_DOWNSTAIRS	0.90	0.93	0.96
WALKING_UPSTAIRS	0.91	0.90	0.94
ACCURACY	0.92	0.93	0.95

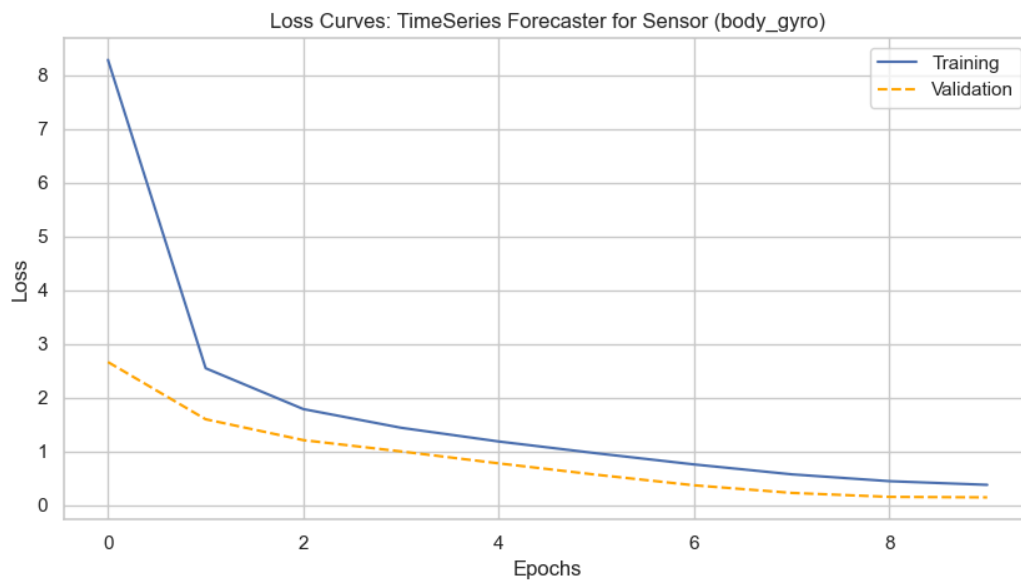
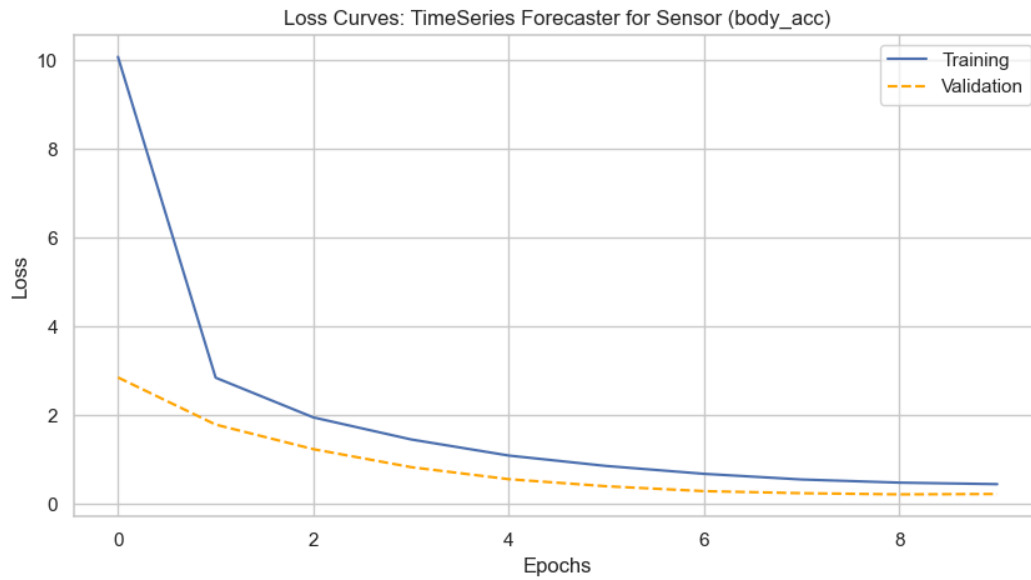
- The Deep Learning Classifiers (LSTM and CNN) outperformed the Random Forest Classifier in F1 and Accuracy Scores. The CNN Classifier performed the best, reaching an accuracy of 95% and very good f1-scores for mobile activities.
- It will be used in our final Dashboard.

4.3.2 Deep Learning Classifiers - Learning Curves



- Both LSTM and CNN Classifiers show similarly small amounts of overfitting occurring after around the 1st epoch.

4.3.3 LSTM Sensor Forecast - Loss Curves



- The LSTM forecaster appears to be learning with no overfitting on both Gyroscope and Accelerometer sensor data. The project was executed with only

10 EPOCHs but based on the above loss curves, the model could benefit from further training.

4.3.4 LSTM Sensor Forecast - Evaluation Metrics

LSTM Forecast Performance on BODY ACC Data

	x-axis	y-axis	z-axis	xyz-Combined
MSE	0.035505	0.018429	0.012027	0.021987
RMSE	0.188427	0.135755	0.109668	0.148281
MAE	0.118676	0.103637	0.083247	0.101853
TraceMSE	2.251556	1.184557	0.740966	1.392360

LSTM Forecast Performance on BODY GYRO Data

	x-axis	y-axis	z-axis	xyz-Combined
MSE	0.170138	0.187729	0.078081	0.145316
RMSE	0.412478	0.433277	0.279430	0.381204
MAE	0.286852	0.327224	0.205940	0.273339
TraceMSE	10.609845	13.079007	5.186324	9.625059

- The LSTM Predictor performs better on Body Acc data than on Body Gyro.
- An overall (xyz-combined) RMSE of 0.148 on Body accelerometer data means the predictions are well within 1 standard deviation of the mean.
- An overall RMSE of 0.381 on Gyro Data is worse but still within 1 standard deviation of the mean.
- Increasing the training time could further reduce these loss metrics.

5. Dashboard Design

5.1 Overview of Tableau Dashboard

- **Visualization 1:** Predicted Live Activity using CNN Classifier

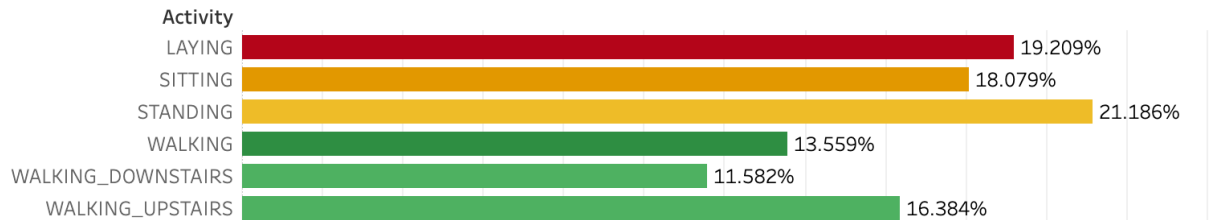


- **Visualization 2:** Amount of time spent continuously the current activity

Good Job for being mobile!
You have been doing this activity for
74.24 seconds now.

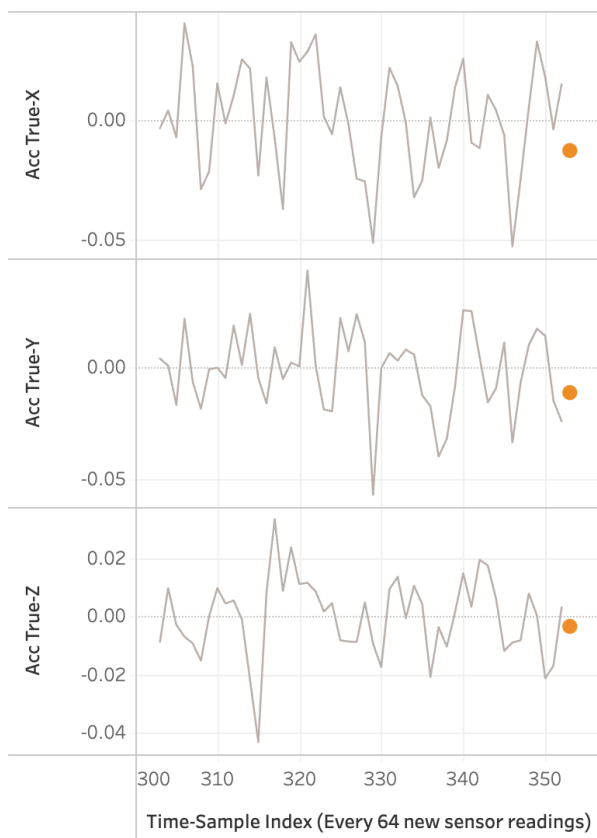
- **Visualization 3: Historical Activity Distribution of the User**

Your historical activity distribution...

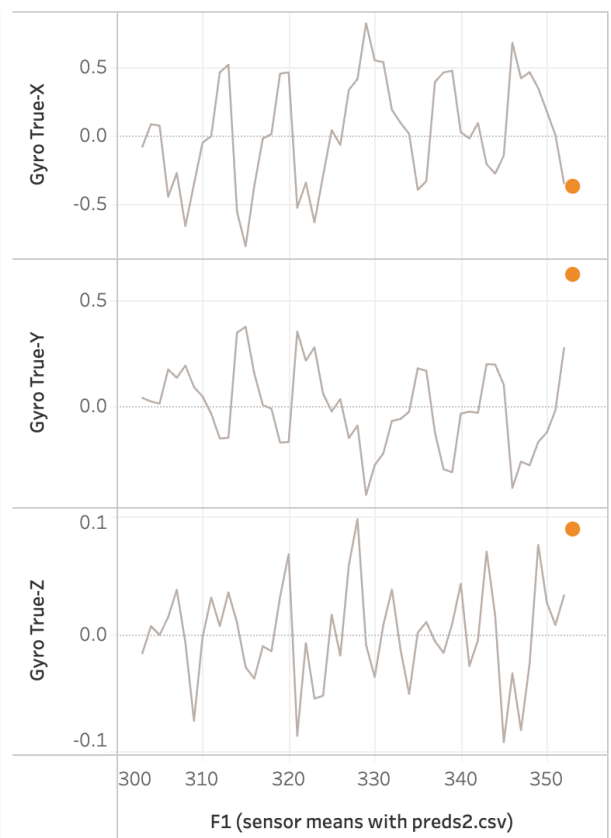


- **Visualization 4: LSTM Forecasts for Body Accelerometer and Body Gyroscope**

Recent Sensor Mean-Readings Next Time Step Forecast (Accelerometer)



Recent Sensor Mean-Readings Next Time Step Forecast (Gyroscope)



5.2 Insights from the Dashboard

- User patterns show **majority sedentary time (sitting/lying/standing) daily.**
- Real-time **classification flags inactivity stretches**, suggesting exercise prompts.
- Sensor reading movement and predictions may be used to alarm users of inactivity. Predictions could be extended to highlight upcoming active periods, **aiding planning.**

6. Discussion and Insights

- **Key Findings:**
 - **CNN hit 95% accuracy** classifying 6 activities from 561 features.
 - **LSTM predicted sensor trends** with 0.14 RMSE on Body Accelerator data, and 0.38 on Body Gyroscope data.
 - **Edge processing** dropped data transmission by 60%, **saving bandwidth.**
- **Practical Implications:**
 - **IoT tracking** can spot inactivity and nudge users toward **fitness goals.**
 - **Scalable** design fits broader health apps.
- **Future Improvements:**
 - Test **transformer models** for **better forecasting** over longer periods.
 - Add text alerts for **real-time inactivity warnings.**

7. Conclusion

This project successfully designed an **IoT-powered health monitoring system**, integrating **smartphone sensors, cloud computing, and machine learning models**.

The system demonstrated **high accuracy in activity classification and future movement prediction**, providing valuable insights into personal health tracking. Future enhancements will focus on **improving real-time analysis and expanding dataset capabilities**.

8. References

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